AUDA-NEPAD
AFRICAN UNION DEVELOPMENT AGENCY

CAMEROON
COUNTRY REPORT ON GENOME EDITING LANDSCAPE ANALYSIS ON AGRICULTURE

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LIST OF ABBREVIATIONS AND ACRONYMS

AFD	Agence Française de Développement
AfPBA	African Plant Breeding Academy
AOCC	African Orphan Crops Consortium
AUDA-NEPAD	African Union Development Agency - New Economic Partnership for Africa's Development
BSL	Biosafety Level
CAADP	Comprehensive Africa Agriculture Development Programme
CARBAP	Centre Africain de Recherches sur Bananiers et Plantains
CBD	Convention on Biological Diversity
GDP	Gross Domestic Product
CEMAC	Communauté Economique et Monétaire de l'Afrique Centrale (Central Africa Economic and Monetary Community)
CIRAD	Centre de coopération Internationale en Recherche Agronomique pour le Développement (French Agricultural Research Centre for International Development)
COP-MOP	Conference of the Parties and Meeting of the Parties
CRID	Centre for Research in Infectious Diseases
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
ECCAS	Economic Community of Central African States
EPA	Environmental Protection Agency
EU	European Union
FAO	Food and Agriculture Organisation of the United Nations
FBPC	Filière Banane Plantain du Cameroun (Banana-Plantain Sector of Cameroon)
GEd	Genome Editing
GMO	Genetically Modified Organism
GTZ	Deutsche Gesellschaft für Technische Zusammenarbeit
IGI	Innovative Genomics Institute
IITA	International Institute of Tropical Agriculture
IIPCC	International Intellectual Property Commercialization Council
IPR	Intellectual Property Right
ISP	Inclusion Support Program
IRAD	l'Institut de Recherche Agricole pour le Développement du Cameroun

MINADER	Ministère de l’Agriculture et du Développement Rural (Ministry of Agriculture and Rural Development)
MINEPDED	Ministère de l’Environnement, de la Protection de la nature et du Développement Durable (Ministry of Environment, Protection OF Nature and Sustainable Development)
MINFI	Ministère des Finances (Ministry of Finance)
MINRESI	Ministère de la Recherche scientifique et de l'Innovation (Ministry of Scientific Research and Innovation)
MINESUP	Ministère de l’Enseignement Supérieur (Ministry of Higher Education)
NAIP	National Agriculture Investment Plan
NBT	New breeding techniques
ODK	Online Data Kit
OAPI	Organisation Africaine de la Propriété Intellectuelle (African Intellectual Property Organization)
OHADA	Organisation pour l'Harmonisation en Afrique du Droit des Affaires. (Organization for the Harmonization of Business Law in Africa)
PCR	Polymerase Chain Reaction
PIISAH	Plan intégré d'import-substitution agropastoral et halieutique Integrated Agro-Pastoral and Fisheries Import-Substitution Plan
SPLR	Supplementary Protocol on Liability and Redress
SPSI	Science–Policy–Society Interface
STEM	Science, Technology, Engineering, and Mathematics
TAAT	Technologies for African Agricultural Transformation
TReND	Teaching and Research in (Neuro) science for Development
TWAS	Third World Academy of Science
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
USAID	United States Agency for International Development
WAITRO	World Association of Industrial and Technological Research Organizations

EXECUTIVE SUMMARY

Background and Purpose: Agricultural biotechnology, particularly genome editing (GEd), is transforming African agriculture by enhancing crop yields, quality, and resilience. This cutting-edge technology is demonstrating significant potential to address food security challenges across the continent, including disease resistance, drought tolerance, nutritional enhancement, and overall crop improvement. The African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) commissioned a Genome Editing (GEd) Landscape Analysis for Cameroon to assess the country's readiness to develop, deploy, and commercialize genome-edited products. The assignment evaluated Cameroon's existing policy, regulatory, institutional, infrastructural, and technical capacities for modern biotechnology and GEd, with a focus on identifying key enablers and constraints across scientific, socio-economic, cultural, and governance dimensions. Specifically, the analysis examined opportunities where GEd can address priority national needs at scale, including agricultural productivity, climate adaptation, food and nutrition security, post-harvest loss reduction, and diversification of healthy diets, as well as identifying staple and indigenous crops with high potential for improving livelihoods. The assessment drew on both secondary data (published literature and institutional databases) and primary data collected through interviews, surveys, and questionnaires. Findings were synthesized across areas such as international treaty commitments, regulatory and policy frameworks, ongoing GEd projects and programs, human and institutional capacity, research and training infrastructure, priority crops and traits, intellectual property considerations, and funding mechanisms.

Key findings

- **Research and development:** Cameroon researchers are actively exploring CRISPR-Cas9-based genome editing for crop improvement. Research teams at the University of Yaoundé I, the University of Buea, and the Institute of Agricultural Research for Development (IRAD) are among the key drivers. Although activities remain at an early stage, progress is evident toward developing varieties with desirable traits such as higher yields, disease resistance, nutritional enhancement, and stress tolerance.
- **Training and human capital:** Capacity-building initiatives are emerging within universities and research centres. However, broader skills development in molecular biology, bioinformatics, and regulatory literacy is still required. To date, no Cameroonian has participated in the specialized African Plant Breeding Academy (AfPBA) CRISPR Course on Genome Editing.

- **Regulatory framework:** Foundational biosafety structures are in place, but a fit-for-purpose regulatory framework for GEd has not yet been established. Clear guidance is needed for research activities, field trials, and potential commercialization.
- **Institutions and private sector:** Government research institutes and universities currently lead early modern biotechnology research and development work. Private-sector involvement remains largely confined to experimental field trials.
- **Priority crops and value chains:** Staple, traditional, and commercial crops that could benefit from genome editing have been identified and in alignment with the National Development Strategy. Priority traits include higher yields, resistance to pests and diseases, drought tolerance, and improved nutritional content.
- **Funding landscape:** Financial support is derived primarily from government line ministries and externally funded competitive grants accessed by researchers through individual or collaborative projects.
- **Intellectual property:** Intellectual property rights in Cameroon's agricultural sector are protected through various legal instruments, including patents, copyrights, and plant variety protection.

Policy implications and recommended actions

Short term (next 12–24 months)

- **Develop genome editing guidelines:** Collaborate with AUDA-NEPAD to provide advisory support and training for key stakeholders to enable the development of a science-based GEd guideline. Existing approved guidelines, policies, and frameworks from other African countries should be reviewed and synthesized to produce a draft tailored to the Cameroonian context and aligned with international best practice.
- **Establish coordination mechanisms:** Create an inter-ministerial genome editing committee involving agriculture, research, economic planning, finance, health, environment, trade, and intellectual property authorities. Mechanisms should also be put in place to ensure the functionality of the National Biosafety Committee.
- **Streamline approvals:** Define clear timelines and institutional roles for permits and trial oversight, and publish a transparent checklist outlining approval processes.
- **Targeted capacity-building:** Fund short courses and laboratory upskilling in genome editing workflows, bioinformatics, quality management, risk assessment, and regulatory compliance.

- **Seed competitive grants:** Launch small grant schemes linking universities, public research institutes, and private actors to accelerate proof-of-concept development.

- **Data consolidation:** Establish a national genome editing registry or portal to reduce information fragmentation and track projects, trials, and outcomes.

Medium to long term (2–5 years)

- **Enact a fit-for-purpose genome editing framework:** Advance development of a GEd regulatory framework within the Biosecurity Project. Activity A.2.1 calls for a review and update of the 2003 National Biosafety Law and its enabling instruments to reflect current global developments, including implementation of the Cartagena Protocol on Biosafety, COP-MOP decisions, and the Supplementary Protocol on Liability and Redress, with specific national rules on liability and redress relating to GMOs.
- **Invest in infrastructure:** Equip core laboratories such as the Biotechnology Centre and Biotechnology Units at the University of Yaoundé I and the University of Buea, and subsequently the University of Ngaoundéré. Additionally, strengthen field trial sites and establish shared facilities with standardized quality assurance and control systems.
- **Foster public–private partnerships:** Incentivize private-sector participation beyond trials to include scale-up, seed systems, and stewardship, and create an enabling environment for investment.
- **Sustainable financing:** Establish multi-year funding mechanisms and facilitate access to regional and international funding opportunities.
- **Strengthen IP and benefit-sharing:** Operationalize plant variety protection systems, licensing templates, and equitable access and benefit-sharing models for locally developed crops, livestock, fisheries, and forestry products.
- **Regional cooperation:** Align national efforts with regional African initiatives to share protocols, training programs, and regulatory best practices.

Conclusion

Genome editing is advancing rapidly and offers significant opportunities to improve crop yields, disease resistance, and climate resilience in Cameroon. While momentum is evident, the technology remains at an early stage of development in the country. With clear and enabling policies and guidelines, targeted investments in human resources and infrastructure, and strong collaboration among stakeholders, genome editing can contribute to climate-resilient, high-yielding crops, improved livestock and fisheries, and a strengthened forestry sector—thereby advancing Cameroon's national food security objectives. Although genome editing offers specific advantages, it should be promoted as part of an integrated bio-innovation approach rather than in isolation.

1. INTRODUCTION

1.1. Description of Cameroon's agricultural landscape

Cameroon's agricultural sector is a vital pillar of the national economy, supporting approximately 70% of the workforce and contributing about 17.29% of the country's GDP and export earnings (O'Neill, 2025). Cameroon is partitioned into five Agro-ecological zones by the Institute of Agricultural Research for Development (IRAD) based on climate, soil, and vegetation, enabling diverse agricultural production. These zones include the Sudano-Sahelian, High Savannah, Highlands, Mono-modal Forest, and Bimodal Forest zones. The Sudano-Sahelian Zone (Far North and North Regions) is characteristic of dry, hot climate with a short rainy season. Major crops grown include sorghum, millet, cotton, cowpea, onion, and groundnut. Guinean High Savannah Zone (Adamawa Region) is characteristic of high altitude and savannah. The major crops grown in the Adamawa region are maize, sorghum, yam, cassava, and soybeans. The Highlands Zone comprise of the West and Northwest Regions and is characteristic of high altitude with temperate climates. The major crops grown are coffee (*Arabica*), potato, maize, beans, plantain, and diverse vegetable farming. The Moomodal Rain Forest Zone comprises of Littoral and South-West Regions and is characteristics of High rainfall, mono-modal climate (one long rainy season). The major crops grown include cocoa, banana, palm trees, rubber, cassava and cocoyam. Bimodal Rain Forest Zone include the Centre, South and East Regions of the country and is characteristic of bimodal rainfall pattern (two rainy seasons). The major crops grown include cocoa, plantain, cassava, maize, and yams (Tang et al. 2022). Agricultural production in Cameroon is dominated by smallholder farmers and cooperatives, alongside a smaller but economically significant agro-industrial sector largely oriented toward export crops.

1.2. National Frameworks and Investments

Cameroon's national agricultural strategy emphasizes structural transformation, productivity enhancement to reduce import dependence, and sustainable development to achieve food security and inclusive economic growth. Key strategic priorities include modernizing agricultural practices, improving rural infrastructure, promoting agro-industrial development, and expanding access to markets and finance. The National Development Strategy (NDS) 2020–2030 (NDS, 2020) identifies agriculture as a central driver of economic transformation, explicitly recognizing that increased agricultural productivity can stimulate industrial growth and improve national food security. Priority agro-industrial value chains include cotton, cocoa, coffee, palm oil, sugar, rubber, rice, maize, banana-plantain, fish, milk, and meat. Cameroon is also addressing climate change impacts on agriculture through initiatives such as the Convergence Initiative and the Science–Policy–Society Interface (SPSI), which promote climate-resilient agriculture and sustainable food system.

The country is aggressively investing in agriculture to modernize its sector, with a 3-year Integrated Agro-Pastoral and Fisheries Import-Substitution Plan (PIISAH 2024-2026) worth CFA1,500 billion with focuses on strengthening rice, corn, milk, and fish value chains, targeting enhanced local production, processing, and marketing. In 2026, the government plans to allocate CFA12.5 billion for the PIISAH programme (Amara, 2026). As of April 2026, the government is focusing on mechanization with subsidies (CFA 400 million) and funding for key sectors like rice, cassava, and palm oil to reduce imports.

1.3. Engagement in Regional Frameworks

Cameroon is actively engaging in regional agricultural frameworks to transform its food systems, enhance food security, and promote sustainable, climate-resilient agriculture. Its efforts are heavily integrated with Central African economic bodies and international partnerships. Cameroon participates in the Economic Community of Central African States (ECCAS) regional agricultural meetings to align its national policies with regional strategies. The Central African Economic and Monetary Community (CEMAC) (Cameroon, Central African Republic, Chad, Congo, Equatorial Guinea, Gabon) have developed a "Common Agricultural Strategy" to address challenges like food insecurity, declining agricultural populations, and the need to increase productivity (Ehode and Ngouna, 2014). While some countries within CEMAC are cautiously moving towards the use of GMOs, others remain cautious. Cameroon is the only country that has authorized the importation of GM maize for consumption, and studies suggest some imported food products (e.g., soy-based oils, children's cereals) in the country are genetically modified (Titanji 2012). However, apart from the advancements in the SODECOTON GM trials, Cameroon remains cautious on modern biotechnology and does not grow GM crops in the fields like in Burkina Faso, South Africa, Nigeria and some other African countries. The region's strategy includes aspects related to the safe and responsible use of GMOs, potentially guided by international protocols like the Cartagena Protocol on Biosafety which has been signed and ratified by almost all members except Equatorial Guinea. There is no regional regulation in place on GEd. The Kampala Declaration's emphasis on food security, sustainability, and resilience closely aligns with Cameroon's agricultural objectives. To advance these goals, the country has developed a comprehensive seed roadmap with support from the Technologies for African Agricultural

Transformation (TAAT) program. This roadmap aims to improve access to high-quality seeds, enhance productivity, and strengthen food security. In alignment with continental agricultural development goals.

1.4. Challenges and Opportunities

Agricultural biotechnology in Cameroon is a critical, emerging sector designed to improve food security for a growing population and boost the economy, which is heavily reliant on agriculture. While the government has aimed to integrate Genetically Modified Organisms (GMOs) into its agricultural strategy to increase productivity, the sector faces significant structural, financial, and regulatory challenges in adopting of biotechnology offers (Ball 2016).

Key opportunities include the following:

- **Increased Productivity and Food Security:** Modern Biotechnology technology offers tools to develop climate-resilient, high-yielding crop varieties (maize, rice, sorghum, cassava) to combat food insecurity and enhance production on less land.
- **Pest and Disease Management:** Partnerships, such as those with Croplife Cameroon, focus on using biotech solutions to fight threats like armyworms and enhance plant disease resistance.
- **Export and Economic Growth:** Strengthening sectors like cocoa and coffee through modern biotechnology can enhance quality and meet international demand, as these constitute over half of the country's export revenues.
- **Research and Development Capacity:** Institutions like IRAD are actively engaged in plant breeding, offering a foundation for broader biotech research.
- **Public-Private Partnerships:** Opportunities exist to collaborate with NGOs and the private sector to improve agricultural entrepreneurship, agricultural insurance, and the adoption of new farming technologies.

Challenges include:

- **Regulatory and Policy Bottlenecks:** Although Cameroon has taken steps towards creating a biosafety framework, legal frameworks for the commercialization of agricultural biotechnology products remain underdeveloped or slow, causing delays in introducing innovative seeds and techniques. Cameroon does not have a GEd guideline.

- **Public perception and ethical concerns:** There is significant debate and scepticism within civil society and among farmers regarding GMO technology. Concerns about human health, environmental impact (such as loss of biodiversity), and the ethical issues surrounding seed sovereignty (dependence on foreign companies) hinder the adoption of biotechnological solutions.
- **Financial and institutional constraints:** Agricultural biotechnology research in Cameroon is primarily driven by the public sector, which has suffered from reduced funding since the economic crisis of the 1980s. There is a lack of sustained, significant investment for biotechnology research and development, leaving research institutions like IRAD with outdated infrastructure and poorly equipped laboratories.
- **Human resources and infrastructure:** There is a shortage of specialized expertise in advanced biotechnology techniques. Furthermore, the lack of modern infrastructure for tissue culture, genetic mapping, GEd and molecular research hinders the development of locally adapted, resilient crop varieties.
- **Weak extension services and technology transfer:** A major gap exists between research institutions and farmers. Traditional farming practices dominate, and there is a lack of capacity to communicate, demonstrate, and distribute biotechnology-driven innovations (like disease-resistant, high-yield plantlets) to small-scale farmers in remote areas.
- **High post-harvest losses and infrastructure deficits:** Beyond biotechnology, the lack of storage facilities and poor transportation infrastructure prevents the benefits of improved production from reaching markets, leading to high post-harvest losses.
- **Climate change and environmental stresses:** While biotechnology offers solutions for climate-resilient crops, the rapid changes in climate (erratic rainfall, drought) in regions like Northern Cameroon add pressure on existing systems, demanding faster, more effective solutions than are currently being delivered.

1.5. Objectives

Against this backdrop, the overall objective of the Genome Editing (GEd) Landscape Analysis was to conduct an in-depth assessment of existing policies, infrastructure, institutional arrangements, financial opportunities, and technical capacities that support product development and commercialization in selected African countries. Specifically, for Cameroon, the Landscape Analysis sought to:

- **Provide an evidence-based description and analysis** of the status of modern biotechnology and genome editing in Cameroon, highlighting key trends, intervening factors, and priority areas for attention. This included assessment of scientific and technical capacity, as well as political, geopolitical, social, human, cultural, and traditional dimensions that either support or constrain the application of genome editing in agriculture and food systems.
- **Identify emerging national needs** that genome editing can readily address, particularly those requiring rapid and scalable solutions. These needs focus on food systems, including agricultural productivity, reduction of postharvest losses, climate adaptation, food and nutrition security, and the promotion of diversified and healthy diets.
- **Identify staple and traditional crops** relevant to Cameroon's national context that have the potential to improve livelihoods through enhanced food security, improved nutrition, climate resilience, and sustainable productivity.

2. APPROACH/METHODOLOGY

Data were collected from both secondary and primary sources. Secondary data (literature review) were gathered from published scientific literature and institutional stakeholder databases, including official websites. Primary data were collected through live interviews using the Online Data Collection Kit (ODK), structured surveys, and email communications based on shared questionnaires prepared in Word document format. In selected cases, online links were used to distribute questionnaires. Primary data were used to validate, complement, and enhance findings from secondary sources. Data from both sources were analysed, synthesized, and organized across the following thematic areas:

2.1 Regulatory and Policy Frameworks

Components of Cameroon's regulatory and policy frameworks for biotechnology, as well as the country's participation in key multilateral environmental agreements, was collected through both secondary and primary sources. This provided insight into the functionality, preparedness, and readiness of Cameroon to adopt and implement GEd technologies in the future since there is no GEd guideline in the country.

2.2 Projects, Crops, Livestock, Fisheries, Forestry, and Traits Ready for Commercialization and Scaling

To understand the status of biotechnology—and particularly genome editing projects, targeted questions were posed regarding ongoing and planned initiatives across crops, livestock, fisheries, and forestry sectors. Sources of funding and key stakeholders involved in these projects were documented using both secondary and primary data sources.

2.3 Traditional and Staple Crops for Genome Editing Improvement

Priority crops with high potential impact were identified based on Cameroon's agricultural development priorities as outlined in the National Development Strategy (NDS 2020–2030). These crops were assessed and have been proposed for improvement of specific traits using genome editing technologies.

2.4 Institutional Capacity (Human Capital, Infrastructure, and Equipment)

During primary data collection, respondents were asked to provide information on existing institutional capacity, including human resources, laboratory and field infrastructure, and equipment relevant to genome editing research, development, commercialization, and scaling. Data were consolidated to generate institution-specific profiles on human capital and infrastructure capacity.

2.5 Stakeholder Mapping

Key stakeholders and institutions, and where appropriate, individual experts, were identified to provide critical information on existing biotechnology and genome editing interventions. Stakeholders were categorized into five (5) groups, as reflected in the data collection tools:

- Regulatory agencies
- Research organizations and institutions.
- Universities
- Private sector and industry
- Government ministries, agencies, and policymakers

2.6 Database Systems and Data Management

Customized data collection tools (questionnaires) and digital platforms were developed to support primary data collection. Questionnaires were tailored to each stakeholder category and used to generate structured datasets that assessed Cameroon's preparedness and capacity to adopt, implement, and scale genome editing technologies.

Enumerators were identified, recruited, and trained through an online induction program on the use of ODK and associated data collection tools. Training was conducted by the IT team and train-the-trainers (ToT) from the Africa Harvest and AGTECH Consortium. All tools and platforms were pre-tested prior to deployment.

Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7 Data Synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and where applicable, tables were generated. Variables investigated included, but were not limited to:

i. signatory to multinational food and environmental agreements;
ii. regulatory framework components;
iii. applications received and approved (field testing, registration, and commercialization);
iv. R&D institutions, personnel, and laboratory/field facilities;
v. projects;
vi. crops and livestock species; and
vii. traits under GEd development.

2.8 Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at: https://genome.africa/

3. RESULTS

3.1 Membership in Multilateral Environment Treaties and Agreements

Cameroon is a signatory to numerous multilateral environmental and food-related treaties and agreements, including the Codex Alimentarius Commission (CAC), signed and ratified in 1969, the United Nations Framework Convention on Climate Change (UNFCCC) signed in 1994 and the Convention on Biological Diversity (CBD) in 1994. In addition, the country ratified the Cartagena Protocol on Biosafety in 2003, and ratified the Nagoya–Kuala Lumpur Supplementary Protocol on Liability and Redress and Access and Benefit Sharing in 2016 as shown in Table 1 below. By ratifying these treaties and agreements, Cameroon commits to global efforts in areas such as food safety, climate action, biodiversity conservation, and environmental protection, and integrates these obligations into its national food safety and environmental management frameworks.

Table 1: Status of Cameroon's participation in key multilateral environmental agreements

Multilateral Environmental Agreements/Treaties	Date of Ratification / Accession by the Country	Reference
Codex Alimentarius Commission is a joint body of the Food and Agriculture Organization (FAO) and the World Health Organization established to develop international food standards, guidelines, and codes of practice. *Critical for risk assessment of food developed through genome editing*	1969 Member	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/
United Nations Framework Convention on Climate, (UNFCCC)	1992 Signed 1994 Ratified	https://unfccc.int/news/cameroon-deposits-instrument-of-ratification-of-paris-agreement
UNEP, RIO Convention on Biological Diversity (CBD)	1994 Ratified	https://www.cbd.int/doc/legal/cbd-en.pdf.
EPA, Cartagena Protocol on Biosafety (CPB)	2001, Signed 2003, Ratified	https://www.cbd.int/doc/legal/cartagena-protocol-en.pdf
UNEP, Nagoya Protocol on Access and Benefit Sharing	2016 Ratified	https://www.cbd.int/abs/nagoya-protocol/signatories

3.2 Laws and Policies that Advance or Hinder Adoption of Modern Biotechnology

National Development Strategy (SND 30) (2020): Cameroon's current national agricultural strategy centres on the National Development Strategy (SND30) for 2020–2030, which operates under the framework of the Cameroon Vision 2035 program. The core focus of this policy is import substitution, food self-sufficiency, and structural agro-industrial transformation to make the nation competitive globally. Cameroon's national agricultural policy integrates biotechnology primarily as a tool for import substitution, food security, and climate resilience, while strictly regulating Genetically Modified Organisms (GMOs) through the national biosafety framework to protect human health, the environment, and local biodiversity

3.3 National Regulatory Framework

3.3.1. Biosafety Act/Law, Biosafety Regulations/Decrees, and GEd Guidelines

Cameroon has three laws (decrees) that promote modern agricultural biotechnology. These decrees are: *The National Framework for Biotechnology in Cameroon (2003), the Legal framework governing biosecurity in Cameroon Law No. 2025/006 of 25 April 2025 and the Legal framework governing biosecurity in Cameroon Law No. 2025/006 of 25 April 2025:*

- **National Framework for Biotechnology in Cameroon (2003)** - The principal legal instrument governing biotechnology in Cameroon is Law No. 2003/006 of 21 April 2003, which establishes the regulatory framework for biotechnology activities in the country. The law establishes a regulatory framework for the safety, development, and use of modern biotechnology. It encourages investment by regulating genetically modified organism (GMO) research, contained use, and cross-border movement, enabling sustainable development in agriculture and health.
- This is complemented with ***Prime ministerial order N°2007/0737/PM (2007****)* which implements the 2003 Biosafety Law, providing practical guidelines for GMO applications.
- **The Cameroon National Laboratory Policy** (2022) provides overarching guidance for laboratory activities nationwide, with the objective of establishing an efficient and coordinated laboratory system by 2030. The policy addresses biosafety, biosecurity, and quality management systems and is supported by complementary guidelines and strategic plans, including a roadmap for quality management system implementation and a national laboratory strategic plan.
- **Legal framework governing biosecurity in Cameroon Law No. 2025/006 of 25 April 2025*:*** This legislation sets mandatory rules for handling genetically modified organisms (GMOs) and protecting biodiversity, aligning national practices with international

biosafety standards. The law encompassing both biosafety and the management of invasive alien species.

The key components of Cameroon’s regulatory framework applicable to modern Biotechnology and GMO products are summarized in Table 2 below. However, Cameroon lacks a dedicated GEd regulatory pathway.

Table 2: Regulatory and institutional landscape for biotechnology products in Cameroon

Institutions	**Mandate / Relevance to GEd**	**Regulatory instruments** (Laws, Acts, Decrees, Regulations, Directives, Guidelines)	**Date of publication**	**Coverage/ scope**	**Reference**
Parliament	Regulations governing Biotechnology in Cameroon	Law No. 2003/006	2003	R&D, Commercialisation and Trade	Cameroon Biosafety Law No. 2003/006. (2003).
	Regulations governing Biosecurity in Cameroon	Law no° 2025/006 of the 25th of April 2025 governing biosecurity in Cameroon	2025		Presidential Decree (2025)
Prime Minister's office	Modalities of application of law governing Biotechnology in Cameroon	Decree nº 2007/737 of 31/05/2007	2007	R&D, Commercialisation and Trade	Prime Minister of Cameroon Order. (2007).
		Order N° 039/CAB/PM of 30th of January 2012 carrying the creation, organization and functioning of the	2012		

		National Biosafety Committee			
Ministry of environment, Protection of nature and sustainable development	Modalities of application of law governing Biotechnology in Cameroon	Decision N° 00334/D/MINEPDED/CAB of 9th of September 2014 constituting the National Biosafety Committee	2014	R&D, Commercialisation and Trade	MIEPDED (2014)
Ministry of Heath	Cameroon national laboratory policy	Cameroon national laboratory policy	2022	R&D, Commercialisation and Trade	Cameroon National Laboratory Policy. (2022).
Prime Minister's Office	Genome Editing Guidelines	None	None	None	

3.3.2 Functionality of the Regulatory Framework

The Ministry of Agriculture and Rural Development (MINADER) has overall responsibility for the agricultural sector; however, its specific mandate in the regulation of genetically modified organisms (GMOs) and GEd products is not clearly defined. Consequently, applications related to the introduction, testing, or release of GMOs or GEd products require clearance from the Ministry of Environment, Protection of Nature and Sustainable Development (MINEPDED).

Together with the Ministry of Scientific Research and Innovation (MINRESI), these ministries share oversight responsibility for the implementation of Cameroon's biotechnology and biosafety legislation. Their mandates cover issues related to recombinant DNA, genetic modification, and genetically modified animals, plants, microorganisms, and viruses. MINRESI oversees national research policy and is responsible for authorizing organizations to conduct research activities in Cameroon. Government institutions engaged in medical and agricultural research operate under the supervision of MINRESI. Cameroon does not have a GEd guideline.

3.4 Socio-Economic Considerations for Decision-Making in Ged Technology and Application

The landscape analysis revealed that socio-economic considerations will play a central role in future decision-making when Cameroon adopts its GEd guideline. These considerations encompass economic outcomes, social structures, cultural practices, and ethical values.

Economic considerations

Key economic impacts identified by respondents include:

- Increased agricultural productivity: Genome editing will have the potential to enhance crop yields, improve farmer incomes, strengthen food security, and reduce import dependence for major commercial crops.
- Market access and trade: The introduction of GEd crops could influence international trade relationships and market access, requiring alignment with trade regulations and export standards.
- Job creation and displacement: While GEd could generate employment opportunities in research, biotechnology, and innovation systems, it may also displace labour associated with traditional farming practices.

- Intellectual property regimes: may affect access to and control over genome editing technologies, particularly for smallholder farmers and public research institutions.

Social impacts

- Public perception and acceptance: Limited public awareness and concerns related to safety, ethics, and perceived risks underscore the importance of inclusive communication and engagement strategies.
- Health and nutrition: Genome editing could offer opportunities to address nutritional deficiencies and crop-related diseases; however, equitable access and risk management are essential.
- Social equity and justice: They emphasized that GEd deployment should avoid exacerbating existing inequalities and ensure benefits are broadly shared across farming communities.

Cultural and ethical considerations

- Respect for traditional practices: Introduction of GEd technologies must respect local cultural norms and traditional farming systems.
- Ethical concerns: Ethical issues related to gene manipulation and unintended consequences were frequently highlighted and require careful consideration.
- Religious beliefs: Religious groups in Cameroon were largely indifferent to the introduction of genome editing technologies.

Decision-making recommendations

- Regulatory frameworks: Cameroon requires a robust, transparent, and science-based regulatory framework for genome editing. Six African countries have already approved GEd guidelines, and AUDA-NEPAD has developed a continental policy framework that could inform Cameroon's approach.
- Public participation: Inclusive public consultation is essential for building trust and ensuring socially acceptable deployment of GEd technologies.
- Capacity building and networking: Investments in human capital, infrastructure, and international collaboration are critical. The study highlighted the need for Cameroonian researchers to engage more actively in international research networks to access expertise, funding, and best practices.

3.5 Factors that Hinder or Advance Application of Biotech and GEd

Political landscape: The Government of Cameroon has expressed growing interest in agricultural biotechnology, including genetically modified organisms (GMOs) and New

Breeding Techniques (NBTs), as part of its broader national agricultural development agenda. The National Development Strategy (SND 2020–2030) aims to restructure the rural economy and promote large-scale, market-oriented agriculture—an objective that some stakeholders interpret as opening space for the adoption of modern biotechnologies.

In 2015, SODECOTON initiated public consultations in the North Region to assess producer perceptions regarding the potential introduction of GM cotton. Following these consultations, open field trials were conducted under the supervision of the National Biosafety Committee and MINEPDED. Outcomes from the research were expected to inform improvements in major cereals such as rice, maize, and sorghum (Mbah, 2018; Mbodiam, 2018).

In 2018, however, SODECOTON discontinued the trials. The company stated that the experimentation was conducted strictly for research purposes and that the experimental phase had concluded in line with authorizations granted by MINEPDED. SODECOTON subsequently shifted focus to the development of high-yielding, non-GM cotton varieties, notably the IRMA Q302 variety, which yields approximately 3.5 tons/ha and is widely cultivated across cotton-growing zones. The company formally indicated that it had no intention of adopting GM cotton to achieve its production objectives (Mbah, 2018; Mbodiam, 2018).

The proposed introduction of GM cotton by the National Cotton Development Company (SODECOTON) generated considerable public debate. Civil society organizations and some researchers raised concerns regarding potential health risks, environmental impacts, and possible effects on the livelihoods of smallholder farmers. Overall, public awareness and understanding of agricultural biotechnology in Cameroon remain limited. Addressing these gaps through targeted communication, education, and public engagement will be critical to building trust and societal acceptance of genome editing technologies. Cameroon remains cautious on modern biotech with no GMO commercialized crop in the country.

Geopolitical factors: Modern biotechnology in Cameroon is significantly shaped by a combination of local agricultural necessities, international regulatory pressure, and the influence of foreign aid and multinational corporations. Key factors include the urgent need for food security, the influence of European standard and external funding. Cameroon government is looking at all the options to increase yields to feed its growing population and some of the options include the use of Genetically Modified Organisms (GMOs) and other modern technologies. Cameroon faces pressure to adopt strict regulations regarding the importation of GMO products, partly influenced by donor nations, including the European Commission, which often opposes or restricts transgenic crops. In addition, most of

Cameroons biotech Research and Development and capacity enhancement programmes are funded by external grants with the researchers compelled to follow their agenda.

Social and human factors: A significant portion of the Cameroonian public fears that modern biotechnology (GMOs) products are not safe, particularly as regards to toxicity or allergies. Critics of the technology also display widespread worry that its resulting products will impact biodiversity negatively by harming the ecosystems, including beneficial insects and indigenous crop varieties. On seed sovereignty, critics argue that the technology will force smallholder's farmers to buy expensive, proprietary seeds from multinational corporate companies every season rather than saving their traditional or landrace varieties.

Culture and traditions: Cameroonians have a passion for "natural/organic" foods that come from their backyard farms which will need public awareness on the introduction of new technologies.

3.6 Genome Editing Programmes and Projects

At present, there are no active genome editing projects in Cameroon. However, historical experience exists with genetically modified crops. In 2012, SODECOTON, in collaboration with Bayer Crop Science, the Institute of Agricultural Research for Development (IRAD), and the Centre for International Cooperation in Agricultural Research for Development (CIRAD), conducted confined field trials of genetically modified cotton.

The Landscape study identified a review paper on CRISPR-Cas–based crop improvement involving two Cameroonian universities (Ntsefong et al., 2023) and a poster presentation applying genome editing approaches to biomedical research (Tsofack, 2023).

3.7 Human Capital and Institutional Capacity

3.7.1 Summary of Human Capital

A small number of Cameroonian scientists have benefited from specialized genome editing and related training programs. These include courses at the Centre for Research in Infectious Diseases (CRID) in Yaoundé, the African Plant Breeding Academy (AfPBA), and international programs delivered by the Innovative Genomics Institute, the African Orphan Crops Consortium, the Seed Biotechnology Center (UC Davis), and the International Institute of Tropical Agriculture (IITA). Additional training has been accessed through the Teaching and Research in Neuroscience for Development (TReND) in Africa program.

Cameroon has over 30 private universities; however, none currently hosts dedicated genome editing research and development programs. Of the eleven public universities distributed across ten regions, several offer relevant coursework within broader programs such as Biochemistry and Molecular Biology, particularly at the Universities of Buea and Yaoundé I. These programs include modules on gene editing techniques, notably CRISPR, within broader genomics curricula.

The study identified several capacity gaps that may constrain future application of genome editing technologies:

- Transformation and regeneration capacity: Limited expertise in transformation and regeneration protocols, especially for underutilized crops.
- Skilled GEd personnel: A small number of trained genome editing researchers, highlighting the need for participation in specialized training programs and complementary skills development in bioinformatics and genomics.
- Collaborative research and networking: Limited integration into international research consortia, which restricts knowledge transfer and access to competitive funding.
- Regulatory clarity: Uncertain or delayed regulatory pathways may contribute to talent loss to countries with more enabling biotechnology environments.

3.7.1.1 Women, Youth and Special Interest Groups Involvement in Modern Biotech and GEd

Women, youth, and special interest groups in Cameroon are increasingly involved in modern biotechnology, primarily within the agricultural, agro-industrial, and health research sectors. Cameroon, is still male-dominated with the proportion of women and youth involvement estimated at 25-27% in the Science, Technology, Engineering, and Mathematics (STEM) fields. This percentage is poised to increase significantly in 2026/7 with the President in his New Year speech assuring the Youth and Women to be prioritized in future job opportunities. In addition, Research and STEM Leadership Initiatives like the Forum des Femmes Scientifiques du Cameroun and the "LES GO SCIENCENT" association are increasing female representation in research. Women scientists are leading studies in parasitology, ecology, and plant biotechnology at institutions like CRID and the University of Yaoundé Biotechnology Centre. H4BF is also a Non-Governmental Organization that empowers women and youth with the tools, skills, and support they need to unlock their full potential, improve their livelihoods, and contribute to their communities' growth and sustainability. In the tissue culture sector, women are increasingly involved in applying tissue culture techniques, specifically in potato production, with IRAD training producers in the use of high-yield potato tissue culture plants in Bayangam. On policy support, the 2026 State Budget prioritizes youth and women-led projects with a significant 50 billion FCFA (https://www.fonajeune.cm/en/)

In the Landscape study, the sample data gathered on key experts in GEd in this Assignment, indicates that 3 out of 12 (approximately 25%) GEd research experts are women, whereas 75% are men. This data wasn't dis-aggregated further to ascertain the fraction that is due to youth and special interest groups like persons with disabilities. To close the gap, government should encourage adaptive training by ensuring specialized training in biotechnology and related sciences is accessible in vocational centres and key state universities.

3.7.2 Research, Development, and Academic Institutions

The Institute of Agricultural Research for Development (IRAD) is the principal public research institution responsible for agricultural research and development in Cameroon. As a public administrative body, IRAD serves as the technical arm of the state in advancing agricultural innovation, often in collaboration with national, regional, and international partners.

IRAD's mandate includes improving livelihoods, diversifying income sources, transferring innovative technologies, and generating intellectual property. The institute operates five Agricultural Research Centres—Maroua, Mbalmayo, Ekona, Bambui, and Wakwa—each aligned with one of Cameroon's agro-ecological zones. These centres have both national and regional mandates to address agricultural development challenges.

Table 4 provides an overview of academic and research institutions in Cameroon with potential capacity to engage in genome editing and related biotechnology activities.

Table 3: Overview of academic and research institutions working on GEd and related capacity in Cameroon.

Institution Name	Dept / Unit	GEd Projects	# of GEd Researchers	Collaborating Partners	Notable Outputs	Gaps Identified
Agricultural Research Institute for Development (IRAD) Nkolbisson-Yaounde	Animal Nutrition	None	1	BecA-ILRI Hub	Capacity building	GEd lab
Agricultural Research Center of Maroua (Far North).	Cereal	None	0	FIDA-PADFA	Cereals (maize, sorghum, millet)	GEd lab
Agricultural Research Centre of Mbalmayo (Centre)	Phyto Pathology	None	1	CORAF	Cocoa Coffee	GEd lab
Agricultural Research Centre of Ekona (Southwest).	Plant Propagation unit	None	1	TWAS	Banana Plantain and legumes	Upgrade of tissue culture lab and a GEd lab
Agricultural Research Centre of Bambui (Northwest).	Plant Propagation unit	None	1	University of Horticultural Sciences of Bagalkot, India and the German Society for International Cooperation (GIZ)1	Potato Common bean	GEd lab

Institution Name	Dept / Unit	GEd Projects	# of GEd Researchers	Collaborating Partners	Notable Outputs	Gaps Identified
Agricultural Research Centre of Wakwa (Adamaoua).	Wheat	None	0	MINEPAT MINADER	Wheat	GEd lab
University of Yaoundé 1	Biochemistry	None	4	Jimma University Ethiopia, Department of Biotechnology, Faculty of Biotechnical Sciences, " Macedonia	Review paper on CRISP-Cas 9	Upgrade of Biotech lab
University of Buea	Biochemistry and Agronomic and Applied Molecular Sciences	None	3	Cameroon Development Corporation, Humbodlt University	plantain	Upgrade of Biochemistry and Applied molecular science laboratories
University of Dschang	Biochemistry	None	1	Department of Microbiology, Synthetic Biology Laboratory, School of Biological Sciences and Biotechnology, Institute of Advanced Research, Koba Institutional Area, Gandhinagar 382007, India	Cereals Potato	Upgrade of Biochemistry laboratory

Institution Name	Dept / Unit	GEd Projects	# of GEd Researchers	Collaborating Partners	Notable Outputs	Gaps Identified
University of Bamenda	Biochemistry Plant Production	None	1	Institut d'Economie Rurale (IER), Bamako, Mali., Univ. Ghana	Maize Sorghum Livestock	Upgrade of Biochemistry laboratory
University of Ngoaundere	Chemistry	None	1	University Yaoundé 1	Sorghum, maize Livestock	Upgrade of Chemistry laboratory
University of Douala	Biochemistry	None	0	Univ Yaoundé 1	Oil palm	Upgrade of Biochemistry laboratory

3.7.3 Training and Professional Development

There are a few Cameroonians who have benefited by participating in plant breeding courses. These researchers could be considered as ambassadors in training and introduction of the technology in their various institutions in the country. Table 5 below highlights the training programs on GEd.

Table 4: Overview of training programmes on GEd

Institution / Organizer	Training Programme	Target Audience / # of Trainees per annual	Frequency	Duration	Gaps Identified
National					
Centre for Research in Infectious Diseases	Functional genomics	Postgraduate (6)	Yearly	5 days	Researchers come back-No lab to practice and implement the technology
International					
Innovative Genomic Institute-IGI, AOCC, AfPBA-UC Davis and IITA	CRISPR Course	Master's and PhD (2)	Yearly	1 year	Researchers come back-No lab to practice and implement the technology
Teaching and Research in (Neuro) science for Development (TReND) in Africa	Genome-editing techniques	Postgraduate (1)	Yearly	5 days	No equipment in laboratories

- **Functional genomics course:** From 24–28 March 2025, a five-day Functional Genomics training course was conducted at the Centre for Research in Infectious Diseases (CRID) in Yaoundé. Participants completed the final module of the training entitled *"Promoter Analysis and Introduction to Gene Editing Using CRISPR-Cas9 Technology."* At the conclusion of the intensive one-week program, the 15 participants reported being significantly empowered by the skills acquired and expressed strong interest in applying the techniques in their respective research activities. Other international courses that Cameroonian researchers have participated include:
- **Innovative genomics institute:** The CRISPR Course is a six-week training program offered by the Innovative Genomics Institute (IGI) in partnership with the African

Orphan Crops Consortium (AOCC), the Seed Biotechnology Centre at the University of California, Davis, and the International Institute of Tropical Agriculture (IITA). The program is delivered under the African Plant Breeding Academy, with training sessions held in Nairobi, Kenya. Three Cameroonian scientists have benefited from the early plant breeding training programs offered through this initiative.

- **TReND courses on genome editing:** Teaching and Research in (Neuro) science for Development (TReND) in Africa is a non-profit organization operated by volunteer scientists from universities worldwide. The organization provides hands-on training to African students and researchers in genome-editing techniques and facilitates the formation of a network of researchers with shared interests in genome editing (https://trendinafrica.org/courses/). Dr. Jacqueline S. Njapdounke Kameni of the University of Ngaoundéré participated in the 2021 TReND genome editing course, held at the University of Ghana and funded by the International Brain Research Organization (IBRO).

In addition, several Cameroon higher institutions of learning, particularly universities, are offering training courses (Certificates, Diplomas, BSc, MSc, and PhD degrees) in biotechnology and its related disciplines, which can be major drivers and leverage for GEd technologies (*See Table 5*).

Table 5: Universities Offering training in Biotechnology and GEd-related courses in Cameroon

Institution	BSc Program	Master's Program	PhD Program
University of Buea, Buea	Biochemistry & Molecular Biology	Molecular Biology & Biotechnology Biochemistry	Biochemistry
University of Yaounde I, Yaounde	Chemistry Biology	Biochemistry	Biochemistry
University of Dschang, Dschang	Biochemistry Plant Production	Biochemistry Plant Biology	Biochemistry
University of Bamenda-Bamenda	Biochemistry	Biochemistry	Biochemistry
University of Ngaoundere, Ngaoundere	Chemistry	Chemistry	Chemistry
University of Douala, Douala	Biochemistry	Biotechnology Diet and Nutrition	Chemistry Biochemistry

3.8 Infrastructure and Equipment

Among the universities and research institutions assessed, six state universities have faculties of agriculture and strong science faculties Biotechnology or Molecular Biology Departments that have the potential of engaging in GEd Research and Development. These institutions possess basic infrastructure, including PCR machines, laminar flow benches, and greenhouses at different containment levels. Generally, most of them lack advanced, high-end research equipment necessary for cutting-edge genome editing work. Collaborative use of facilities and shared infrastructure could enhance research outputs.

In 1986, IRAD established a biotechnology laboratory in Ekona, with support from USAID, focusing on in vitro culture techniques to produce yam and cocoyam plantlets. Subsequently, a laboratory and experimental station was established in Njombé in collaboration with CIRAD to support banana plantlet production.

During the same period, the Biotechnology Centre (BTC) at Nkolbisson, University of Yaoundé (now University of Yaoundé I), was established with funding from the Government of Cameroon and international partners including ISP, USAID, and the European Union. The centre focused on recombinant DNA and hybridoma technologies for research on tropical diseases such as onchocerciasis, malaria, and tuberculosis. The plant science unit conducted in vitro culture work on cocoyam but did not engage in genetic modification. Researchers at the centre also developed rhizobial and mycorrhizal biofertilizers for environmentally sustainable agriculture.

In the early 2000s, oil palm seedlings were produced using conventional breeding techniques at the IRAD station in Dizangue (now Dibamba) in the Littoral Region.

In 2004, the Biotechnology Unit at the University of Buea established protocols for the detection of genetically modified food and feed, with funding from MINEPDED. The unit was subsequently accredited as a Reference Laboratory for GMO testing. Additional GMO detection laboratories have since been established at IRAD Yaoundé, IITA Nkolbisson, and the Biotechnology Centre at the University of Yaoundé I. Société Générale de Surveillance (SGS) also operates a GMO testing laboratory in Douala.

The International Institute of Tropical Agriculture (IITA) conducts biotechnology research in Cameroon focused on disease diagnostics using PCR techniques and field trials, particularly for cassava, cocoa, and banana. However, most advanced genome editing research by IITA is conducted at its research stations in Kenya and Tanzania.

Table 6 presents the status and needs assessment of biosafety laboratory facilities across institutions in Cameroon. Overall, most institutions possess approximately 60–75% of the equipment listed in Annex A, highlighting both existing capacity and critical gaps.

Despite the availability of basic infrastructure, the studies identified significant limitations related to funding constraints and a shortage of genome-editing experts in most institutions. In addition, procurement challenges—particularly those related to the importation of specialised equipment and dependence on sole suppliers—were noted as factors that can delay research activities, although streamlined importation processes could help accelerate procurement timelines.

Importantly, the assessment revealed that three institutions are fully equipped to support advanced genome editing research and development. These institutions are well positioned to serve as hubs of excellence and could collaborate through consortia or shared-access models to support less-resourced institutions, accelerate product development, and optimize national investments in the future genome editing infrastructure.

Table 6: Status and needs assessment of biosafety laboratory facilities by institution.

Institution	**Type of Facility**	**Biosafety Level (or 2)**	**Status/ Equipped (see Annex A)**	**Limitations**	**Support Needed**
University of Buea	Lab / greenhouse/ field trials	1 & 2	Not fully	Funding maintenance	Lab chemicals and equipment
University of Yaounde 1	laboratory	1 & 2	Not fully	Funding maintenance	Lab chemicals and equipment
University of Dschang	Lab / greenhouse/ field trials	1	Not fully	Funding maintenance	Lab chemicals and equipment
University of Bamenda	Lab / greenhouse/ field trials	1	Not fully	Funding maintenance	Lab chemicals and equipment

University of Ngoaundere	Lab/greenhouse	1	Not fully	Funding Maintenance Electricity	Lab chemicals and equipment
University of Douala	Laboratory	1	Not fully	Funding Maintenance	Lab chemicals and equipment
IRAD-Centres Cameroon	Lab / greenhouse/ field trials	1	Not fully	Funding Maintenance	Lab chemicals and equipment
IITA-Cameroon	Lab/greenhouse/field	1 & 2	Not fully	None	Lab chemicals and equipment

3.9 Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/Breeds for Improvement Using GEd

Globally, genome editing is increasingly being applied to a wide range of crops, primarily to improve yields and enhance tolerance to biotic and abiotic stresses. These applications aim to boost agricultural productivity, reduce hunger, and address nutritional deficiencies by improving crop nutrient profiles (Tiwari et al., 2021). As GEd technologies become more accessible and cost-effective, they offer substantial benefits for consumers—such as improved nutrition, enhanced food safety, and reduced food waste—as well as for farmers, through increased resistance to pests, diseases, and weeds, and reduced seed production costs. Genome editing also presents opportunities to enhance climate resilience and biodiversity within cropping systems (Ceasar & Kavas, 2024). Genome editing in crops, livestock and fisheries has the potential to address several critical challenges facing Cameroon's animal production systems. GEd technologies can be applied to enhance disease resistance, productivity traits, and animal welfare. CRISPR-based tools also offer opportunities for rapid and precise disease diagnostics, which could strengthen early detection, prevention, and control of livestock and aquaculture disease outbreaks.

Productivity-related applications include improvements in growth rate and feed efficiency in cattle, sheep, goats, and poultry. Genome editing may also be used to modify milk composition to better meet nutritional needs, such as increasing beneficial proteins (e.g., lactoferrin) or reducing allergens (e.g., beta-lactoglobulin). Additionally, GEd could be applied to improve meat quality, particularly in Cameroon's cattle industry, where certain breeds are associated with tougher meat textures.

Environmental stressors present another opportunity for GEd application. Heat stress significantly affects livestock productivity in many parts of Cameroon, especially in the northern regions. Genome editing could contribute to the development of heat-tolerant breeds of cattle, sheep, and goats. Despite these opportunities, the study found that genome editing has not yet been applied in livestock or fisheries research in Cameroon.

Genome editing also holds significant potential for advancing agroforestry systems in Cameroon by improving tree crop productivity. Although the technology remains relatively new in the country, ongoing discussions and exploratory efforts suggest growing interest in understanding its applicability and addressing associated technical, regulatory, and ethical challenges. There are currently no GEd projects in the country (Table7).

3.9.1 Priority Crops for Genome Editing in Cameroon.

The priority crops are selected based on the countries National Development Strategy (NDS) 2020–2030 (NDS, 2020) and the Cameroon Agricultural Investment Plan 2020-2030 (MINADER, 2020) and the National Institute of Statistics (NIS) (

Cocoa: Over 500,000 Cameroonians depend on cocoa production for their livelihoods. Cameroon is Africa's fourth-largest producer of cocoa and ranks among the world's top five producers of quality cocoa. Cameroonian cocoa beans are distinguished by their darker colour, reddish break, and unique pungent flavour, making them particularly attractive to the European cocoa-pressing industry due to their high cocoa butter content and antioxidant polyphenol levels (International Trade Centre, 2001; ONCC, 2024). Genome editing could be used to enhance these desirable quality traits and address major production constraints, particularly black pod disease caused by *Phytophthora palmivora* (Ndoumbe-Nkeng et al., 2004).

Banana and plantain: Banana exports declined from 19,866 tons in 2023 to 16,004 tons in 2024, highlighting production challenges in the sector. Plantain remains a major staple crop in Cameroon. A significant cost driver in banana and plantain production is the control of black Sigatoka leaf spot disease, caused by *Mycosphaerella fijiensis* (Etebu & Young-Harry, 2001). Genome editing could offer environmentally friendly and cost-effective solutions to manage this disease and stabilize yields.

Cassava: Cassava production in Cameroon is projected to increase from 5.5 million to 10 million tons annually, with yields rising from 25 to 30 tons per hectare. However, pests and diseases remain major constraints. Genome editing could be instrumental in improving disease resistance, yield stability, and post-harvest quality, thereby unlocking the crop's full potential.

Cereal crops: Staple grains such as maize, sorghum, and millet could benefit from GEd applications targeting drought tolerance, enhanced nutritional content, and resistance to pests and diseases, especially in the country's semi-arid and northern regions.

Livestock: Cameroon has a vibrant livestock industry. Genome editing technology could contribute to improving resistance to major livestock diseases such as foot-and-mouth disease, as well as enhanced productivity and resilience in cattle and poultry systems.

Based on the National Development Strategy (SND 2020–2030) and the Cameroon Agricultural Investment Plan 2020-2030 (MINADER, 2020), priority staple, traditional, and commercial crops, livestock, forestry, and fisheries species requiring improvement through genome editing are presented in Table 6. The identified traits include enhanced productivity, reduced post-harvest losses, improved climate adaptation to drought, pests and diseases, improved nutritional quality, diversified and healthy diets, and socio-economic importance. Improving the productivity and resilience of these priority commodities will contribute directly to Cameroon's food sovereignty and help reduce the country's growing food import bill.

Table 7: National priority crops that could potentially benefit from GEd technology

Crops/Livestock/ Agroforestry/Fisheries	**Trait improved of interest**	**Socio-Economic Justification**	**GEd Potential (Low/Medium/High)**	**Existing R&D**	**Actual Annual Production Capacity (tonnes)**	**Expected Annual Production Capacity (tonnes)**
Cereals						
Sorghum	Striga, draught stress tolerance and nutritional enhancement	Climate smart staple food for human/ animal feed and beverages for small holder farmers and key ingredients in the local beer industry,	High	Agronomy	1.2 million FAO (2023)	1.5 million
Maize	Viruses/ weeds resistance	Major staple food crop for humans and animals. Raw material for various industries	High	Agronomy	2.2 million Trading Economics (2026)	2.8 million
Millet	Draught Stress, nutrition enhancement	Climate smart food security and income generation for small holder farmers	Medium	Agronomy	1,1 million	100,000
Legumes						
Cowpea	Pests and Disease resistance	Key source of food, income, and livestock feed, particularly for small-scale farmers. It is a vital source of protein, particularly for the poor,	Medium	Agronomy	31,000 *Tchiagam et al. 2024*	40,000
Bambara Groundnut	Pests and Disease resistance	Drought-resistant legume that provides a vital source of food and income.	Low	Agronomy	40,640 *Majola et al. 2021*	42,000
Fruit crops						

Crops/Livestock/ Agroforestry/Fisheries	Trait improved of interest	Socio-Economic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual Annual Production Capacity (tonnes)	Expected Annual Production Capacity (tonnes)
Banana and plantains	Pests and Diseases - Black sigatoka Banana Streak Virus	Major daily staple crop in every household and banana major export earner. 150,000 ha planted	High	Agronomy	5.4 million plantains 1.2 million bananas Tamfu (2026)	10 million Plantain. 1,5 million bananas
Tree crops						
Cocoa	Viruses and flavour Cocoa swollen shoot Virus	500,000 Cameroonians are dependent on cocoa production for their livelihoods. Fourth exporter. 1.2% to the national GDP and 8.2% to the agricultural GDP	High	Agronomy	300,000 *ONCC, 2024*	600,000
Coffee	Diseases and pests' resistance	Special premium Arabica coffee with high demand in Europe	Medium	Agronomy	34 000 Cameroon Fact Sheet (2025)	160,000
Tuber crops						
Cassava	Virus resistance	Major daily staple food, 16th worldwide production; a source of income for farmers, and a raw material for various industries.	High	Agronomy	5.5 million Guardian Post (2023)	10 million
Cocoyam *Macabo*	Diseases and pests' resistance	Staple food crop, 2nd to Nigeria who is No 1 in the world, a source of income for farmers,	Medium	Agronomy	2.0 million Onyeka (2014)	2.5 million

Crops/Livestock/ Agroforestry/Fisheries	Trait improved of interest	Socio-Economic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual Annual Production Capacity (tonnes)	Expected Annual Production Capacity (tonnes)
Taro		and a food security measure, particularly for those in rural areas.				
Cotton	Pests and diseases resistance	73% of the region's total production in 2021. The industry employs over 220,000 farmers, primarily in the northern regions	High	Agronomy	350,000	400,000
Livestock						
Cattle	Foot and mouth resistance, higher milk production	Provides a substantial portion of the country's meat and milk supply, contributes to the GDP, and is a source of income for farmers and pastoralists	High	Breeding	1.2 million herds MINEPIA (2023)	3 million herds
Chicken	Disease resistance	Food security, income generation, and social well-being. Provides a source of animal protein, especially for poorer populations.	High	Breeding	55,000 MINEPIA (2023)	80,000
Fish	Disease resistance	Food security, livelihoods, and vital source of animal protein,	High	Breeding	241 000 MINEPIA (2023)	300,000

Source: National Development Strategy (NDS) 2020–2030; Cameroon Agricultural Investment Plan 2020-2030; National Institute of Statistics (NIS)

3.10 Intellectual Property Rights and Benefit Sharing

Several national, regional, and international institutions play a role in governing intellectual property rights (IPR) in Cameroon. At the regional level, the African Intellectual Property Organization (OAPI) serves as the primary intergovernmental body regulating intellectual property across its Francophone member states, including Cameroon. Cameroon is also a member of the World Intellectual Property Organization (WIPO), which supports the protection of intellectual creations globally. In addition, the International Intellectual Property Commercialization Council (IIPCC) maintains a local chapter in Cameroon, with a mandate focused on intellectual property commercialization and technology transfer.

Intellectual property rights in Cameroon's agricultural sector are protected through several legal instruments, including patents, copyrights, and plant variety protection (PVP)**.** These instruments are designed to safeguard innovations in plant breeding, agricultural technologies, and associated knowledge systems, ensuring that inventors and innovators can benefit from their work. Patents/Plant Breeders Protection may be granted for novel agricultural technologies and genetic traits, conferring exclusive rights to inventors for a defined period.

For genome-edited crops, intellectual property protection is primarily governed by the Bangui Agreement of March 2, 1977**,** which underpins the regional intellectual property framework administered by OAPI. This agreement provides the legal basis for patents and plant variety protection across member states and would apply to genome-edited agricultural innovations once such products enter the research, development, or commercialization pipeline.

To date, Cameroon has not managed any intellectual property cases related to genome editing or modern agricultural biotechnology. The only initiative that could have triggered IP management considerations was the SODECOTON–GM cotton project, which was discontinued at the confined field trial stage in 2018. As a result, no patents, licenses, or benefit-sharing arrangements related to GEd technologies have been developed or operationalized in the country. A summary of the IPR legal instruments and organisations is presented in Table 8.

Table 8: IPRs, legal instruments (laws) and organizations in charge

Country	Intellectual Property Rights	Organization / Institution	Laws / Regulations / Protocols / Treaties
Cameroon	Patents, Copyrights, Trade names and Plant Variety Protection (PVP)	African Intellectual Property Organization (OAPI)	Bangui Agreement of March 2, 1977

	Global IPRs issues	World Intellectual Property Organization (WIPO)	The Paris and Berne Conventions (1967)
	IPRs commercialization	International Intellectual Property Commercialization Council (IIPC)	International treaties, regional regulations, and domestic laws.

3.11 Private Sector Participation

The private sector plays a critical role in innovation ecosystems by driving research and development, partnerships, technology licensing, and commercialization of genome-edited products. In many contexts, this involves close collaboration with public research institutions to navigate intellectual property regimes, regulatory frameworks, and the specific needs of smallholder-dominated farming systems.

In Cameroon, however, private-sector participation in biotechnology and genome editing remains limited. As noted earlier, the only authorization to cultivate a genetically modified crop (GM cotton) was granted to a para-public company SODECOTON. SODECOTON carried this trial together with the private sector company Bayer Crop Science and two public research institutions IRAD and CIRAD. Most private companies in Cameroon's agricultural sector currently rely on research and development conducted by parent companies or partners based outside the country. They carry out field trials for local adaptation and distribution.

Despite this limited engagement, Cameroon has a growing number of small and medium-sized enterprises (SMEs) in agriculture that are open to adopting innovative technologies. These include companies such as, AgriTech, Cresus Agro, SGS Sustainable Oils Cameroon, the Cameroon Development Corporation (CDC), and PAMOL.

Cameroon also has well-established commodity-specific associations and parastatal organizations that influence production decisions and value-chain governance. These include the National Cocoa and Coffee Board and SODECAO (cocoa and coffee); SOCAPALM (oil palm); the National Association of Actors of the Banana–Plantain Sector of Cameroon (FBPC); SODECOTON (cotton); the Association des Opérateurs de la Filière Production Sucre du Cameroun and the Cameroon Sugar Corporation (sugarcane); and regional rice corporations in the North, North-West, and South-West regions. These entities play a central role in shaping crop priorities and could serve as entry points for future GEd adoption.

The private seed sector in Cameroon primarily focuses on cereals and vegetable seeds. Locally based companies include Grenier du Monde Rural, Semences de Qualité Garantie, and Farmers' House, which specialize in traditional and locally adapted species. International seed companies operating in Cameroon include MAÏSCAM, PANAAR, Seed Co, Bayer, Corteva, and Semagri. Notably, Bayer and Corteva have been involved mainly in communication and advocacy related to modern biotechnology regulation, rather than direct genome editing research or commercialization. An overview of the private sector mostly seed companies involved in modern Biotechnology is presented in Table 9.

Table 9: Overview of private sector companies involved in seed production and distribution, crop improvement, and biotech/GEd activities in Cameroon

Company	Type (Multinational / Regional / National	Biotech/GEd Activities	Partnerships	Challenges Faced	Investment Interest
Grenier du Monde Rural	National	No Biotech/GEd	IRAD, FASA	Regulatory, Accessibility, Climate change; Finance to buy hybrid seeds.	Distribution and Crop breeding (conventional)
Semences de Qualité Garantie	National	No Biotech/GEd	IRAD, CARBAP,	Certification, Climate change, Counterfeit market	Crop Breeding
Farmers House	National	No Biotech/GEd	IRAD	Regulatory, Supply chain, Climate change.	Distributor
MAÏSCAM (Maïserie du Cameroun)	National	No Biotech/GEd	IRAD, Société Anonyme des Brasseries du Cameroun	Climate change, Volatile pricing	Importation and Local purchase
Seedco	Multinational	No Biotech/GEd	IRAD and IITA,	Regulatory, Informal seed markets, Climate change	Crop breeding (conventional)
Corteva	Multinational	No biotech/ GEd in Cameroon	IIAM	Regulatory, Informal seed markets, Climate change	Crop breeding (conventional)
Bayer	Multinational	Biotech/No GEd in Cameroon	IITA, IRAD, SODECOTON	Regulatory, Counterfeit Markets, Climate change	Crop breeding (conventional)
Syngenta	Multinational	No Biotech/GEd in Cameroon	IRAD, IITA, CropLife	Regulatory, Counterfeit Markets, Climate change	Crop breeding (conventional)

3.11 Funding and Investment Landscape

Funding for higher education and agricultural research in Cameroon is largely derived from government budget allocations. Universities receive funding primarily through the Ministry of Higher Education (MINESUP), while national research institutions are funded through the Ministry of Scientific Research and Innovation (MINRESI). The Ministry of Agriculture and Rural Development (MINADER) also provides targeted funding to the Institute of Agricultural Research for Development (IRAD) for priority crops and value chains.

In addition to domestic public funding, researchers in Cameroon access competitive international grants to supplement limited national resources. These external funding sources, summarized in Table 10, are critical for sustaining research activities, including those related to biotechnology and plant breeding.

It remains difficult to determine the exact proportion of national expenditure dedicated specifically to biotechnology. However, available evidence indicates that Cameroon’s agricultural research spending remains low. Plant breeding programs—closely linked to biotechnology—have been affected by economic constraints and limited public investment. In 2014, agricultural research spending accounted for approximately 0.34% of agricultural GDP, significantly below the 1% target recommended under continental and international frameworks. This persistent funding gap constrains the country’s ability to develop, test, and scale advanced technologies such as genome editing.

Table 10: Overview of national and other potential funding sources for genome editing (GEd)

Funder/Donor	**Organization Type**	**GEd Project**	**Amount (USD)**	**Duration**	**Recipient Institution(s)**	**Area of Focus**
Ministry of Finance, MINESUP, MINRESI, MINADER and MINEPAT	Government	None	None	Yearly	All government universities and IRAD	Ag Biotech
European Union	Development funding	None	None	Yearly	All government universities and IRAD	Ag Biotech
World Association of Industrial and Technological Research Organizations (WAITRO)	Development funding	None	None	Yearly	All government universities and IRAD	Ag Biotech
United States Agency for International Development (USAID)	Development funding	None	None	Yearly	All government universities and IRAD	Ag Biotech
Deutsche Gesellschaft für Technische Zusammenarbeit (GIZ)	Development funding	None	None	Yearly	All government universities and IRAD	Ag Biotech
Third World Academy of Sciences (TWAS)	Development funding	None	None	Yearly	All government universities and IRAD	Ag Biotech
GEF	Development funding	None	381 462,58	5 years (1st phase) 3 years (2nd phase)	Biotechnology Unit of the University of Buea & Biotechnology Center of the University of Yaounde 1	Biodiversity Biosafety
MINEPDED	National funding (cofinance)	None	423 750	5 years (1st phase)	Biotechnology Unit of the University of Buea &	Environmental protection

Funder/Donor	Organization Type	GEd Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
				3 years (2nd phase)	Biotechnology Centre of the University of Yaounde 1	

3.12 Stakeholders Involved in Biotechnology

Several key stakeholders were identified and mapped in Cameroon to obtain critical primary data on the adoption of GEd technologies as shown in Annexure 2. The stakeholders have been grouped into different categories namely:

Regulatory and other government agencies: These MINADER, MINEPDED, MINRESI; and the Ministry of Livestock, Fisheries and Animal Industries (MINEPIA). These institutions play central roles in policy formulation, regulation, oversight, and implementation of biotechnology-related activities.

Universities and teaching institutions: Academic institutions involved in biotechnology education and research include the University of Buea, University of Yaoundé I, University of Bamenda, University of Dschang, University of Douala, and the University of Ngaoundéré. These institutions contribute to capacity building, research, and human resource development in modern biotechnology and related disciplines.

National agricultural research institutes and CGIAR centres: Key research organizations include IRAD and its specialized centres, as well as the International Institute of Tropical Agriculture (IITA). These institutions are actively involved in agricultural research, crop improvement, and technology development relevant to biotechnology and genome editing.

Private sector: Private sector stakeholders include agro-industrial companies, seed companies, and multinational biotechnology firms such as Cameroon Development Corporation (CDC), PAMOL, SODECOTON, SEMRY, SOCAPALM, Corteva Agriscience, Bayer, Syngenta, BASF, Grenier du Monde Rural, Semences de Qualité Garantie, MAÏSCAM, PANAAR, and Seed Co. These actors are important for technology deployment, seed production, commercialization, and advocacy for modern agricultural technologies.

Governmental and non-governmental organizations: Relevant government and civil society actors include MINADER, MINEPDED, the Ministry of Economy, Planning and Regional Development, the Ministry of Forestry and Wildlife, and MINEPIA, as well as non-governmental organizations involved in agriculture, environment, and biosafety-related activities.

These stakeholders were engaged through their institutional heads during the collection of primary data. Data collection methods included live interviews conducted using ODK platform, as well as structured surveys administered through emailed questionnaires. A detailed list of the stakeholders consulted is provided in the Annexure 2.

4. RECOMMENDATIONS AND POLICY OPTIONS

Genome editing is advancing rapidly and presents significant opportunities to improve crop productivity, nutritional quality, resistance to pests and diseases, and climate resilience. Realizing these benefits requires an early and deliberate commitment to a regulatory structure that ensures food and environmental safety while protecting intellectual property rights for both publicly and privately developed biotechnology products. This must be complemented by strong research institutions equipped with adequate human capital and infrastructure to generate locally relevant innovations and communicate the value of genome-edited crops to producers. As adoption progresses, farmers gain practical experience with the technology, leading to productivity gains and improved livelihoods.

A comprehensive policy approach for genome editing in Cameroon therefore requires both short-term and long-term strategies. Short-term actions should prioritize establishing an appropriate regulatory framework, strengthening research and regulatory capacity, and engaging key stakeholders. Long-term objectives should focus on fostering innovation, ensuring the responsible and ethical use of the technology, and maximizing its socio-economic and environmental benefits while mitigating potential risks.

Based on the findings of the landscape studies, the following recommendations are proposed to ensure that Cameroon effectively harnesses the potential of genome editing technologies:

- **Regulatory framework:** Cameroon should develop a robust and fit-for-purpose regulatory framework for genome editing that governs research, development, and application of GEd technologies while ensuring safety, transparency, and meaningful public engagement. To date, at least six African countries have approved national genome editing guidelines, and AUDA-NEPAD has developed a continental policy framework for the application of GEd. These existing instruments can serve as valuable reference points for the development of Cameroon’s national GEd guidelines.
- **Capacity building:** Targeted investment in training and education is essential to build national capacity in genome editing research, development, and regulation. Cameroonian researchers should be supported to participate in short- and long-term GEd-specific training programs and international plant breeding courses. A growing number of Cameroonian scientists already trained in these programs can spearhead the initiation of genome editing research within their home institutions.
- **Infrastructure and equipment:** Most of the selected universities and research centres possess basic molecular biology laboratories that require upgrading to support

genome editing research and innovation. Prioritizing the upgrade of laboratories at the University of Buea and the University of Yaoundé I into dedicated GEd research facilities would significantly accelerate national capacity development.

- **Funding for strategic GEd programs:** Dedicated funding mechanisms should be established to support strategic genome editing projects and programs that address national agricultural and food system challenges. This could be achieved through innovative and targeted funding initiatives led by MINADER and MINEPAT.
- **Private sector participation:** Private sector engagement should be strengthened by promoting start-up incubation programs in agricultural biotechnology and genome editing, and by encouraging the establishment of biotechnology companies through public–private partnerships. Universities and research institutions should collaborate closely with the private sector to enhance innovation, technology transfer, and competitiveness.
- **Networking and collaboration:** Cameroonian researchers should actively build national, regional, and international networks and collaborate with global experts in genome editing and related fields. The formation of research consortia will enhance access to large-scale funding opportunities, advanced training, and shared research infrastructure.
- **Public participation and consultation:** Inclusive public engagement and consultation are critical for building trust, addressing concerns, and ensuring that genome editing technologies are developed and deployed in a socially acceptable and beneficial manner. Transparent communication strategies should be integrated into policy and regulatory processes.
- **Focus on traditional and staple systems:** Genome editing efforts should prioritize traditional and staple crops, livestock, agroforestry species, and fisheries that are critical to food security, nutrition, and livelihoods in Cameroon. Priority species and value chains suitable for improvement through GEd have been identified in this study and should guide research and investment decisions.

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ANNEXES

Annex 1: Criteria for Determining Laboratory Status for Bsl-1 And Bsl-2 Operations

(i.) **Criteria for infrastructure and equipment for BSL 1:** 3-4 rooms containing the following: PCR, Incubator, Centrifuge, Sanger and Next Generation Sequencers, Freezers (-80, -20), P/ATC room, Access to consumables, Laminar Air Flow chamber, Electrophoresis Apparatus, balances and scales, Autoclave, Freezers, Microwave, Vortexer, UV illuminator

(ii.) **Criteria for infrastructure and equipment for BSL 2:** Standard Microbial Practices + Special practices + All BSL-1 equipment plus a mandatory biosafety hazard sign, special protective gear, special Cabinets (class II), controlled access to rooms etc., handling agents of moderate potential hazards to people + animals + environment

	Conditions	**Status**
BSL 1	If all in (i) above are available with or without the sequencer	Fully equipped
	Missing any of the other equipment in addition to the sequencer	Not fully equipped
BSL 2	Conformance to the criteria in (ii) above	Fully equipped
	Any non-conformance to the criteria in (ii) above	Not fully equipped

Annex 2. List of institutions and resource persons involved in Biotechnology research and product development in Cameroon.

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SN	NAME	SECTOR	MINISTRY/DEPARTMENT/INSTITUTION/ ORGANIZATION	POSITION	CONTACT DETAILS
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